

Module 04: Disk Scheduling — SCAN and C-SCAN

1. Comparison: SCAN vs C-SCAN

Parameter	SCAN (Elevator)	C-SCAN (Circular SCAN)
Service Direction	Both directions (Reverses at boundary)	Strictly ONE direction only
Return Path	Serves requests on way back	Immediate fly-back to 0 without service
Wait Time Variance	Higher variance (favors middle cylinders)	Uniform wait time for all cylinders
Boundary Touch	Touches 0 or MAX cylinder	Touches both 0 and MAX cylinders

Disk Scheduling: SCAN (Elevator) vs C-SCAN (Circular)

Queue: 98, 183, 37, 122, 14, 124, 65, 67 | Head = 53 | Direction: Towards 0

SCAN (Elevator Algorithm)

• Rule: Disk arm ek direction me aakhiri end (0) tak jata hai, phir reverse hokar doosre end ki taraf badhta hai.

Path: 53 → 37 → 14 → 0 → 65 → 67 → 98 → 122 → 124 → 183

Calculations:

- Downward to End 0: (53 - 0) = 53 tracks
- Reverse from 0 to 183: (183 - 0) = 183 tracks

Total Movement = 53 + 183 = 236 Cylinders

Total Head Movement = 236 Cylinders

• Advantage: No starvation! (Like an elevator)

• Inequity: Middle cylinders get serviced twice as often as boundary cylinders.

C-SCAN (Circular SCAN)

• Rule: Arm sirf EK DIRECTION me requests serve karta hai. End par pahunch kar sidha bina service kiye start par aata hai.

Path (Moving Up): 53 → ... → 199 → 0 → 14 → 37

Calculations (Upwards direction):

- 53 to 199: (199 - 53) = 146 tracks
- Fly-back 199 to 0: (199 - 0) = 199 tracks (No service)
- 0 to 37: (37 - 0) = 37 tracks

Total Movement = 146 + 199 + 37 = 382 Cylinders

• Advantage: Uniform waiting time for all cylinders!

• Treats cylinders as a continuous circular ring.

2. Solved Numerical Result

- **SCAN (towards 0):** (53 - 0) + (183 - 0) = 236 Cylinders.
- **C-SCAN (towards 199):** (199 - 53) + (199 - 0) + (37 - 0) = 382 Cylinders.